PRODUCT REQUIREMENTS DOCUMENT (PRD) — RELEASE V1.0.0

Tactical Grid Duel

1v1 Turn-Based Server-Authoritative Grid Arena Game

Author: Lead System Architect & Engineering Team | Date: October 2026 | Status: Production Ready

Project Metadata	Specification Details
Application Type	Full-Stack Real-Time Multiplayer Web Application
Frontend Architecture	React 18 + Vite + Tailwind CSS + Lucide Icons + Socket.io-Client
Backend Architecture	Node.js 20+ / Express 5 + Socket.io 4 (Persistent WebSockets)
Database Infrastructure	Hybrid: SQLite (WAL mode, game.db) / Supabase Cloud PostgreSQL
Deployment Targets	Vercel (Client Edge Network) + Render.com (WebSocket Web Service)
Security / Anti-Cheat	100% Server-Authoritative Game Mechanics & Turn Validation

1. Executive Summary & Product Vision

Tactical Grid Duel is a competitive, high-stakes 1v1 turn-based grid arena game designed for instant web browser execution across mobile devices and desktop computers. Traditional browser-based games frequently suffer from client-side tampering, state desynchronization, high latency, or cumbersome onboarding. Tactical Grid Duel solves these issues through a 100% server-authoritative architecture where all movements, Action Point (AP) deductions, damage calculations, and vector knockback physics are verified by a centralized Node.js engine before broadcast.

- **Core Value Proposition:** Accessible zero-install PvP tactical arena with zero-lag WebSocket synchronization and seamless cross-device compatibility.
- **Anti-Cheat Guarantee:** The browser client functions purely as a dumb terminal / presentation layer; all game states and timers reside authoritatively on the server.

2. Target Audience & User Personas

- **The Competitive Strategist (Alex, 24):** Values quick tactical depth, calculated risk vs. reward, and zero-cheat transparency in a competitive environment.
- **The Casual Browser Duelist (Sam, 19):** Seeks instant 3-minute battles against friends across mobile and desktop without account creation friction.

3. Core Game Mechanics & Rules Engine

The game environment consists of a discrete 5x5 matrix (coordinates X: 0-4, Y: 0-4). Two commanders are deployed in diagonally opposing corners and compete to eliminate the opponent.

Metric / Rule	Specification	Server-Side Rule Enforcement
Arena Bounds	5x5 Grid (Tiles A1 to E5)	Coordinate clamp 0 <= X <= 4 and 0 <= Y <= 4.
Starting Positions	P1: (0, 0) | P2: (4, 4)	Auto-assigned upon room initialization.
Health Points (HP)	10 Max HP per player	Target HP <= 0 triggers terminal win condition.
Action Points (AP)	3 AP replenished per turn	Non-accumulative; unspent AP expires at turn end.
Movement Cost	1 AP per step	Validates Manhattan Distance (|dx| + |dy| == 1).
Strike Attack Cost	2 AP per strike	Deals 2 DMG + calculates vector knockback.
Knockback Vector	Delta direction: (tx - x, ty - y)	Pushes target 1 cell away or wall-slams.
Turn Time Limit	30 Seconds strict	Server interval auto-switches turn on timeout.

VECTOR KNOCKBACK SPECIFICATION: Knockback Physics Formula: When Actor A at (x1, y1) attacks Actor B at (x2, y2), the knockback destination is calculated as: target = (x2 + (x2 - x1), y2 + (y2 - y1)). If target is outside the 5x5 grid, the knockback slams into the wall without displacement.

4. Functional Requirements

FR-1: Matchmaking & Room Management

Users can initiate a Quick Match or enter a custom Room Code. The server maintains in-memory room objects with a strict 2-player capacity limit. The duel commences automatically upon the arrival of the second player.

FR-2: Server-Authoritative Movement & AP Consumption

Upon receiving a move request, the server verifies active turn status, AP balance (>= 1 AP), and adjacent cell legality. If valid, the position is updated, AP is decremented, and the updated state is broadcast to the room.

FR-3: Combat & Lethal Damage Calculation

Adjacent enemy targeting triggers an attack. The server deducts 2 AP, subtracts 2 HP from the victim, and computes the directional knockback. If enemy HP reaches 0, the game transitions to "ended" and awards the victory.

FR-4: Disconnection & Forfeiture Handling

If an active player abruptly disconnects during gameplay, the server automatically declares the remaining player the winner by forfeit and commits the outcome to the persistent database.

FR-5: Real-Time Combat Feed

Every verified action generates an immutable event log entry (movement, strike, timeout, knockback) broadcast to all room participants and rendered in a real-time terminal feed.

5. System Architecture & Network Topology

Layer / Tier	Technology	Hosting / Runtime	Operational Purpose
Client (UI)	React 18 + Vite	Vercel Edge Network	Reactive UI, tactical grid rendering, sound/animations
Network Protocol	Socket.io 4 + HTTP	TLS / WSS over port 443	Bi-directional duplex messaging under 50ms latency
Application Tier	Node.js + Express 5	Render.com (Web Service)	Authoritative game state machine, timers, room memory
Persistence Tier	Supabase Cloud (PostgreSQL)	AWS Cloud Managed	Global ranking tables, historical match storage
Local Fallback	SQLite (WAL Mode)	Local Disk (game.db)	Zero-network local development and offline testing.

Real-Time Socket Event Protocol Specification

Event Name	Direction	Payload Signature	Description
join_game	Client -> Server	{ username, roomId }	Requests placement into an arena room
assigned_role	Server -> Client	{ role: "p1"|"p2", roomId }	Confirms role assignment to user socket
game_state	Server -> Client	{ players, turn, turnTimeLeft, ... }	Authoritative state broadcast to all room peers
make_move	Client -> Server	{ targetX, targetY }	Dispatches move or strike action to server
end_turn	Client -> Server	void	Manually concludes turn and resets AP
timer_tick	Server -> Client	{ turnTimeLeft, turn }	Synchronizes 1-second authoritative timer
game_error	Server -> Client	{ message: string }	Emitted when an invalid/illegal move is attempted
game_over	Server -> Client	{ winner, loser, turns, reason }	Terminates match and shows victory modal.

6. Data Model & Database Schema Specification

The database supports dual engines: Cloud PostgreSQL (via Supabase) for live production and SQLite (via better-sqlite3 / sql.js) for local development.

Table: players

Field Name	Data Type	Constraints	Description
id	BIGSERIAL / INTEGER	PRIMARY KEY	Unique player surrogate identifier.
username	TEXT / VARCHAR	UNIQUE NOT NULL	Unique commander callsign.
wins	INTEGER	DEFAULT 0	Total cumulative victories.
losses	INTEGER	DEFAULT 0	Total cumulative defeats.

Table: matches

Field Name	Data Type	Constraints	Description

id	BIGSERIAL / INTEGER	PRIMARY KEY	Unique match record identifier.
winner	TEXT	NOT NULL	Username of winning commander.
loser	TEXT	NOT NULL	Username of defeated commander.
turns	INTEGER	NOT NULL	Total turns elapsed before match conclusion.
played_at	TIMESTAMP / DATETIME	DEFAULT NOW()	Exact timestamp of match completion.

7. Non-Functional Requirements (NFR)

- **NFR-1: Latency & Response Budget:** Round-trip time for validated moves must not exceed 60ms over standard broadband or 4G LTE connections.
- **NFR-2: Security & Zero-Trust Client Model:** No game parameter (HP, AP, position, timer) can be mutated via client payload. Server performs 100% parameter validation.
- **NFR-3: Cross-Device Responsiveness:** Fully responsive viewports across standard mobile smartphones (375px+), tablets, and widescreen desktop monitors (1920x1080).
- **NFR-4: Fault Tolerance & Reconnection:** Graceful room cleanup upon socket dropouts; automatic database retry with exponential backoff on network blips.

8. UI/UX Design & Layout Architecture

The client follows a high-contrast Cyber-Tactical dark aesthetic utilizing Tailwind CSS color palettes:

- **Lobby Interface:** Minimalist deployment panel with Commander Name input, optional Room Code field, Quick Match button, rules cheat sheet, and live SQLite/Supabase leaderboard preview.
- **Left Panel (Status & Controls):** Player 1 and Player 2 cards displaying animated HP bars, 3 AP diamond indicators, coordinates, turn-active glowing halos, and manual End Turn button.
- **Center Panel (5x5 Arena Grid):** Tactical grid with alphanumeric coordinates (A-E, 1-5). Dynamic tile colorization: Emerald green hover for legal movement (1 AP), Rose red hover for adjacent enemy strike (2 AP). Top bar features big countdown timer and copyable room code.
- **Right Panel (Telemetry & Rankings):** Real-time combat history feed with color-coded events (movement, attacks, knockbacks, timeouts) and searchable live database leaderboard.
- **Game Over Modal:** Full-screen backdrop blur modal rendering Victory Trophy or Defeat Skull, match statistics, and one-click "Play Again" or "Return to Lobby" triggers.

9. Product Evolution & Future Roadmap

Version Milestone	Target Capabilities	Status / Release Target
v1.0.0 (Current)	Core 1v1 Arena, Server-Authoritative Engine, Supabase/SQLite Hybrid, Vercel/Render Deploy	Production Released
v1.1.0	Destructible terrain cover, randomized obstacle walls, and energy hazards on grid	Planned Q1 2027
v1.2.0	Solo AI Commander mode using Minimax with Alpha-Beta pruning for offline play	Planned Q2 2027
v1.3.0	Web Audio API procedural sound effects, haptic vibration on mobile, and particle trails	Planned Q2 2027
v2.0.0	Ranked ELO competitive matchmaking ladder, commander classes, and custom abilities.	Planned Q3 2027

> **VERIFICATION & DEPLOYMENT STATUS:** Tactical Grid Duel is verified and released for production deployment across Vercel (Client) and Render (Server) with active Supabase cloud persistence.